



Exercise 1

Linear Block Codes

Goal of the exercise : Finding good codes with large lengths is an important research topic in coding theory. We will construct linear block codes of length $2n$ starting from two linear block codes of length n .

1. Consider the single parity-check code of length 4. Find a systematic generator matrix and a systematic parity-check matrix for this code. Find also the minimum distance of the code.
2. Consider the repetition code of length 4. Find a systematic generator matrix and a systematic parity-check matrix for this code. Find also the minimum distance of the code.

We compare two different constructions to obtain a code of length $2n$ from two codes of length n . Consider the codes

$$\mathcal{C}_1(n, k_1, d_1) \quad \text{and} \quad \mathcal{C}_2(n, k_2, d_2).$$

Naïf Construction. Under this construction the new code is obtained by concatenating each codeword of $\mathcal{C}_2$ with each codeword of $\mathcal{C}_1$:

$$\mathcal{C}_{\text{naïve}} = \{c = (u|v) : u \in \mathcal{C}_1, v \in \mathcal{C}_2\}.$$

It can be verified that $\mathcal{C}_{\text{naïve}}$ is linear whenever $\mathcal{C}_1$ and $\mathcal{C}_2$ are linear.

3. Find the dimension $k_{\text{naïve}}$ of $\mathcal{C}_{\text{naïve}}$.
4. Let G_1 and G_2 be generator matrices for codes $\mathcal{C}_1$ and $\mathcal{C}_2$. Using the matrices G_1 and G_2 , find a generator matrix $G_{\text{naïve}}$ for code $\mathcal{C}_{\text{naïve}}$.
5. Show that the minimum distance of $\mathcal{C}_{\text{naïve}}$ is $d_{\text{naïve}} = \min\{d_1, d_2\}$.
6. Show that for any code $\mathcal{C}_{\text{naïve}}(2n, k_{\text{naïve}}, d_{\text{naïve}})$:

$$2d_{\text{naïve}} \leq 2n - k_{\text{naïve}} + 2. \quad (1)$$

Squared construction (also called « $u | u + v$ » construction). The new code is obtained as follows :

$$\mathcal{C}_{\text{carré}} = \{c = (u|u+v) : u \in \mathcal{C}_1, v \in \mathcal{C}_2\}.$$

It can be verified that the so obtained code $\mathcal{C}_{\text{carré}}$ is linear.

Many good codes are based on the squared construction, including the family of popular Reed-Müller codes.

7. Recall that G_1 and G_2 denote generator matrices of codes $\mathcal{C}_1$ and $\mathcal{C}_2$. Using G_1 and G_2 , find a generator matrix $G_{\text{carré}}$ of the code $\mathcal{C}_{\text{carré}}$. What is the dimension $k_{\text{carré}}$ of $\mathcal{C}_{\text{carré}}$?
8. Show that the minimum distance of code $\mathcal{C}_{\text{carré}}$ is given by $d_{\text{carré}} = \min\{2d_1, d_2\}$.
We suggest that you distinguish the following three cases in your proof : $(u = 0, v \neq 0)$, $(u \neq 0, v = 0)$ and $(u \neq 0, v \neq 0)$.
9. Apply the squared construction to $\mathcal{C}_1$ being a single parity-check code of length 4 and $\mathcal{C}_2$ being a repetition code of length 4. Find length, dimension, and minimum distance of the resulting code $\mathcal{C}_{\text{carré}}$. Find also a generator matrix for $\mathcal{C}_{\text{carré}}$.
10. Is it possible to find a code $\mathcal{C}_{\text{naïve}}$ with same length, dimension, and minimum distance as the code $\mathcal{C}_{\text{carré}}$ found in 9. ? ($\mathcal{C}_{\text{naïve}}$ does not necessarily have to be constructed from the same codes $\mathcal{C}_1$ and $\mathcal{C}_2$ as $\mathcal{C}_{\text{carré}}$.)